Chapter 13. Classification of Covering Spaces

Equivalence of coverings, universal covering space, covering transformations, existence of coverings

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Throughout, $p : E \to B$ is a covering map with E and B path connected and locally path connected, $p(e_0) = b_0$. We classify such coverings by subgroups of $\pi_1(B, b_0)$.

13.1 Equivalence of covering spaces

Coverings $p : E \to B$ and $p' : E' \to B$ are **equivalent** if there is a homeomorphism $h : E \to E'$ with $p' \circ h = p$.

Lemma 13.1 (General lifting lemma)

Let Y be path connected and locally path connected and $f : (Y, y_0) \to (B, b_0)$ continuous. Then f lifts to $\tilde{f} : (Y, y_0) \to (E, e_0)$ iff $f_*\pi_1(Y, y_0) \subset p_*\pi_1(E, e_0)$; the lifting is unique.

Proof. Necessity is $f_* = p_*\tilde{f}_*$. For sufficiency define $\tilde{f}(y)$ as the endpoint of the lift (from e_0) of $f \circ \alpha$, where α is any path from y_0 to y; the subgroup condition makes it independent of α (the loop $f \circ (\alpha * \bar{\beta})$ lifts to a loop). Continuity uses local path connectedness of Y and evenly covered neighbourhoods. Uniqueness: the set where two liftings agree is open and closed. ■

Theorem 13.2

p and p' are equivalent by an h with $h(e_0) = e_0'$ iff $H_0 = p_*\pi_1(E, e_0)$ and $H_0' = p_*'\pi_1(E', e_0')$ are equal. They are equivalent (with some base points) iff H_0 and H_0' are conjugate in $\pi_1(B, b_0)$.

Proof. Lift p through p' and p' through p (Lemma 13.1); the compositions are liftings of p, p' fixing base points, hence identities. Changing e_0 within the fibre along a path $\tilde{\alpha}$ conjugates H_0 by $[p \circ \tilde{\alpha}]$, and every conjugate arises. ■

13.2 The universal covering space

A covering $q : Y \to B$ with Y simply connected is a **universal covering**. By Theorem 13.2 it is unique up to equivalence, and it covers every other covering: if $p : E \to B$ is a covering, there is a covering $r : Y \to E$ with $p \circ r = q$.

Example 13.1

$\mathbb{R} \to S^1$, $\mathbb{R}^2 \to T^2$, $S^2 \to P^2$, and $\mathbb{R}^2$ for every T_n $(n \geq 1)$ and P_m $(m \geq 2)$ (hyperbolic geometry for $n \geq 2$, $m \geq 3$). The universal covering of the figure eight is the infinite 4-valent tree.

13.3 Covering transformations

A **covering transformation** (deck transformation) is an equivalence $h : E \to E$ of p with itself; they form a group $\mathcal{C}(E, p, B)$. Let $H_0 = p_*\pi_1(E, e_0)$ and $N(H_0)$ its normalizer.

Theorem 13.3

$\mathcal{C}(E, p, B) \cong N(H_0)/H_0$. If H_0 is normal (the covering is **regular**), then $\mathcal{C}$ acts transitively on each fibre and $\mathcal{C} \cong \pi_1(B, b_0)/H_0$; in particular for the universal covering $\mathcal{C} \cong \pi_1(B, b_0)$.

Proof. For $e_1 \in p^{-1}(b_0)$ there is a deck transformation taking e_0 to e_1 iff $p_*\pi_1(E, e_1) = H_0$, i.e. iff the class $[\alpha]$ corresponding to e_1 under the lifting correspondence lies in $N(H_0)$; the map $[\alpha] \mapsto$

(the deck transformation sending e_0 to e_1) is a surjective homomorphism $N(H_0) \to \mathcal{C}$ with kernel H_0. ■

Conversely, if a group G acts on a path connected, locally path connected X **properly discontinuously** (every x has a neighbourhood U with $gU \cap U = \emptyset$ for $g \neq 1$), then $X \to X/G$ is a regular covering with deck group G, and $\pi_1(X/G)/p_*\pi_1(X) \cong G$. For simply connected X, $\pi_1(X/G) \cong G$: e.g. lens spaces $S^3/(\mathbb{Z}/n)$, and $\mathbb{R}^2/\mathbb{Z}^2 = T^2$.

13.4 Existence of covering spaces

B is **semilocally simply connected** if every point has a neighbourhood U such that loops in U are trivial in $\pi_1(B)$. This is necessary for a universal covering to exist.

Theorem 13.4

Let B be path connected, locally path connected and semilocally simply connected. For every subgroup $H \subset \pi_1(B, b_0)$ there is a covering $p : E \to B$ with $p_*\pi_1(E, e_0) = H$. Hence the equivalence classes of connected coverings correspond bijectively to conjugacy classes of subgroups of $\pi_1(B, b_0)$.

Construction. Let E be the set of equivalence classes of paths α in B starting at b_0, where $\alpha \sim \beta$ if they end at the same point and $[\alpha * \bar{\beta}] \in H$; let $p(\alpha^{\#}) = \alpha(1)$. Topologize E by the sets $B(U, \alpha) = \{(\alpha * \delta)^{\#} : \delta \text{ a path in } U \text{ from } \alpha(1)\}$, U a path connected neighbourhood. Semilocal simple connectivity makes p a homeomorphism on each $B(U, \alpha)$ for suitable U, and these sets are disjoint for inequivalent α over the same U, so p is a covering. The lift of a loop f ends at $f^{\#}$, which equals the base point iff $[f] \in H$; so $p_*\pi_1(E) = H$. ■

The **Hawaiian earring** (circles of radii $1/n$ tangent at a point) is not semilocally simply connected and has no universal covering.